

Probability & statistics for ML

A refresh, applied to the problems Revolut cares about — fraud, credit, disputes, A/B tests. Read once, then answer the questions at the end **out loud, in English**.

WHAT THIS MODULE IS FOR

The “ML basis: maths + statistics” round. Not to make you a statistician — to make sure that when they say “*the model is overconfident*”, “*the lift isn't significant*” or “*precision at that recall*”, you know exactly what they mean and can say it back with a number.

Rule for the whole programme: every concept gets three things — the idea in plain words, the formula (so a symbol never scares you), and one Revolut-shaped example. Nothing else.

00 Notation refresh — the symbols you'll see

Ten minutes. If you can read these, you can read any formula in the interview.

SYMBOL	READ IT AS	EXAMPLE
Σ	“add up”	Σx_i for $i = 1 \dots n = x_1 + x_2 + \dots + x_n$. $(1/n) \Sigma x_i$ is the mean.
Π	“multiply together”	Probability of independent events happening all together.
$\log$	“how many times do I multiply the base to get this”	$\log$ turns products into sums: $\log(a \cdot b) = \log a + \log b$. ML uses it constantly because probabilities multiplied get tiny; their logs add nicely. Natural log (base $e \approx 2.718$) unless stated.
e^x / $\exp(x)$	the inverse of log	$\exp(\log x) = x$. Shows up in the sigmoid and in growth.
$P(A)$	probability of A	Between 0 and 1. $P(\text{fraud}) = 0.002 = 0.2\%$ of transactions.
$P(A \mid B)$	probability of A given B	$P(\text{fraud} \mid \text{new device})$: fraud rate among transactions from a new device.
$E[X]$	expected value = long-run average	Expected loss per transaction.
$\text{Var}(X)$, σ^2 · σ	variance · standard deviation	How spread out values are. σ is in the same units as X.
$\mathbf{x}$ (bold) / $\mathbf{x} \in \mathbb{R}^d$	a vector of d numbers	An embedding: a customer as a list of, say, 256 numbers.
$\mathbf{w} \cdot \mathbf{x}$ or $\mathbf{w}^T \mathbf{x}$	dot product	$w_1 x_1 + w_2 x_2 + \dots$: a weighted sum. Every linear model is this.
argmax / argmin	“the value that makes this biggest / smallest”	Training = argmin of the loss over the weights.
$\hat{y}$ (y-hat)	the model's prediction of y	y is the truth (defaulted or not), $\hat{y}$ the prediction.

01 Probability — the four rules you need

1 • Probability is a long-run frequency

$P(A)$ = the fraction of times A happens if you repeat the situation many times. Always between 0 and 1; all outcomes together sum to 1.

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If 2 of every 1,000 card transactions are fraudulent, $P(\text{fraud}) = 0.002$. This number — the **base rate** — is the single most important number in any fraud or default problem, and the one people forget.

2 • Independence and the multiplication rule

Two events are **independent** if one tells you nothing about the other. Then $P(A \text{ and } B) = P(A) \cdot P(B)$. If they are *not* independent: $P(A \text{ and } B) = P(A) \cdot P(B | A)$.

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"New device" and "fraud" are *not* independent — fraud is much more common on new devices. That dependence is exactly what a model learns. Treating dependent things as independent is the classic naive mistake (it's literally why "Naive Bayes" is called naive).

3 • Conditional probability

$$P(A | B) = P(A \text{ and } B) / P(B)$$

"Among the cases where B is true, how often is A true." Every model output is a conditional probability:

$$P(\text{default} | \text{what we know about this customer})$$

4 • Bayes' theorem — updating a belief with evidence

$$P(A | B) = P(B | A) \cdot P(A) / P(B)$$

In words: *what I believed before* ($P(A)$, the prior) $\times$ *how well the evidence fits* ($P(B|A)$), scaled by *how common the evidence is overall* ($P(B)$).

REVOLUT — THE EXAMPLE TO KNOW BY HEART

A fraud model flags a transaction. The model catches 90 % of fraud ($P(\text{flag} | \text{fraud}) = 0.9$) and wrongly flags 2 % of legitimate transactions ($P(\text{flag} | \text{legit}) = 0.02$). Fraud base rate 0.2 %. **If a transaction is flagged, what's the probability it's actually fraud?**

$$\begin{aligned} P(\text{flag}) &= 0.9 \cdot 0.002 + 0.02 \cdot 0.998 = 0.0018 + 0.01996 = 0.02176 \\ P(\text{fraud} | \text{flag}) &= 0.9 \cdot 0.002 / 0.02176 \approx 0.083 \end{aligned}$$

Only ~8 %. A "90 % accurate" detector is wrong 92 % of the time it fires, because fraud is rare and the 2 % false-alarm rate on a huge legit population swamps it. This is why **precision** (next module) and the **base rate** matter more than "accuracy" in fraud, AML and credit. If you can explain this on a whiteboard, you've passed half the round.

02 Random variables, expectation, variance

A **random variable** is a number whose value depends on chance: the amount of a transaction, whether a customer defaults (0/1), the number of disputes a merchant gets this month.

Expected value — the long-run average

$$E[X] = \sum x_i \cdot P(x_i) \text{ (discrete)} \quad E[X] = \int x \cdot p(x) dx \text{ (continuous)}$$

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Expected loss on a €1,000 loan with 4 % default probability and 60 % loss if default:

$E[\text{loss}] = 0.04 \cdot 0.6 \cdot 1000 = €24$. That's the number the interest rate has to cover — exactly what your lender model was for: getting $P(\text{default})$ per customer so the price per customer is right.

Variance and standard deviation — how spread out

$$\text{Var}(X) = E[(X - E[X])^2] \quad \sigma = \sqrt{\text{Var}(X)}$$

σ is in the same units as X , so it's the one you quote. Roughly: for many well-behaved variables, ~68 % of values fall within $\pm 1\sigma$ of the mean, ~95 % within $\pm 2\sigma$.

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A customer's typical spend is €40 $\pm$ €25 (mean $\pm$ σ). A €300 transaction is more than 10σ away — “unusual for this customer”. That z-score idea, $z = (x - \text{mean}) / \sigma$, is the simplest anomaly detector, and it's what PRAGMA does in a much richer way: “how far is this event from what this customer normally does”.

Covariance and correlation — do two things move together

$$\text{corr}(X, Y) = \text{Cov}(X, Y) / (\sigma_x \sigma_y) \text{ between } -1 \text{ and } +1$$

Correlation $\neq$ causation — say it before they do. And correlation only measures *linear* relationships; two variables can be strongly related with correlation ≈ 0 .

03 Distributions — the five you'll meet

DISTRIBUTION	MODELS	REVOLUT EXAMPLE	KEY FACTS
Bernoulli	one yes/no trial with probability p	Does this customer default? Is this transaction fraud?	mean p , variance $p(1-p)$. The target of every binary classifier.
Binomial	number of “yes” in n independent trials	How many of 1,000 loans default?	mean np , variance $np(1-p)$. Basis of A/B test maths.
Poisson	count of events in a time window when events are rare and independent	Disputes per merchant per month; logins per hour	mean = variance = λ . If variance $\gg$ mean, something's clustered (e.g. a fraud ring).
Normal (Gaussian)	a symmetric bell; sums/averages of many things	Sample means of anything (see CLT); model errors, often	68 / 95 / 99.7 % within 1 / 2 / 3 σ . <i>Transaction amounts are NOT normal</i> — they're heavy-tailed.

DISTRIBUTION	MODELS	REVOLUT EXAMPLE	KEY FACTS
Log-normal / heavy-tailed	positive, skewed, rare huge values	Transaction amounts, customer balances, LTV	Mean is dragged up by whales; quote the median and percentiles. Take the log before modelling.

THE TRAP THEY LIKE

"Average transaction is €85." With a heavy tail that can mean most people spend €20 and a few spend €5,000. Always ask: mean or median? What does the 95th percentile look like? Saying that in the interview signals you've handled real data.

04 Sampling, the Central Limit Theorem, confidence intervals

Why you can trust an average from a sample

Central Limit Theorem (CLT): take a sample of n values from almost any distribution and compute the mean. Do it many times: the means are normally distributed around the true mean, with standard deviation

$$\text{standard error} = \sigma / \sqrt{n}$$

Two consequences you must be able to say: (1) the sample mean is a good estimate of the true mean; (2) **precision improves with $\sqrt{n}$, not n** — to halve the error you need *four* times the data.

Confidence interval

$$95\% \text{ CI} \approx \text{mean} \pm 1.96 \cdot \sigma / \sqrt{n}$$

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You measure a 30 % dispute auto-resolution rate on 400 cases. For a proportion,

$\sigma = \sqrt{p(1-p)} = \sqrt{0.3 \cdot 0.7} \approx 0.46$, so $SE = 0.46 / \sqrt{400} = 0.023$, and the 95 % CI is **30 % $\pm$ 4.5 %**.

If someone asks "is it 30 or 32?", the honest answer is "somewhere between 25 and 35 — we need more cases to say more."

Bias vs. noise in sampling

More data fixes **noise** (random error). It does *not* fix **bias** (systematic error: your sample isn't like the population). Classic ML biases: *survivorship* (you only see loans you approved — you never learn what the rejected ones would have done), *selection* (the pilot ran on your best customers), *label leakage* (the training data contains the answer). Naming survivorship bias in a credit context is worth a lot in this round.

05 Hypothesis tests, p-values, A/B tests

The logic

You want to know if a change had an effect. You assume it didn't (the **null hypothesis, H_0**) and ask: *if H_0 were true, how surprising is the data I got?* That surprise is the **p-value**: the probability of seeing a result at least this extreme *if there is no real effect*. Small p (conventionally < 0.05) $\rightarrow$ reject $H_0 \rightarrow$ "statistically significant".

What a p-value is NOT: it is not the probability that H_0 is true, and it is not the size of the effect. A tiny effect on a huge sample is "significant" and useless; a big effect on a small sample is "not significant" and

maybe real. Always ask for the **effect size and its confidence interval**, not just the p.

The two errors

	REALITY: NO EFFECT	REALITY: REAL EFFECT
You say "effect"	Type I error (false positive), probability α (usually 5 %)	correct — "power" (usually target 80 %)
You say "no effect"	correct	Type II error (false negative), probability β

Same table as a fraud model's false positives / false negatives — it's the same maths. **Power** is the chance you detect a real effect; it depends on effect size, sample size and α . Under-powered tests are the #1 A/B mistake.

A/B test, end to end

1. **Metric** first (one primary, a few guardrails), and the **minimum effect worth detecting** (e.g. +1 pp on resolution rate).
2. **Sample size** from that: roughly $n \text{ per group} \approx 16 \cdot \sigma^2 / \delta^2$ for 80 % power at $\alpha = 5 \%$ (δ = the effect you want to detect). For a proportion, $\sigma^2 = p(1-p)$.
3. **Randomise** properly (by customer, not by session), run for full weekly cycles, **don't peek and stop early** — peeking inflates false positives.
4. Read: effect size, CI, p. Check guardrails. Then decide.

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Baseline auto-resolution 30 %; you want to detect a lift to 32 % ($\delta = 0.02$). $\sigma^2 = 0.3 \cdot 0.7 = 0.21$.
 $n \approx 16 \cdot 0.21 / 0.0004 = 8,400 \text{ per group}$. If disputes run at 2,000/day, that's ~8–9 days for both arms. If someone proposes "let's test on 300 cases", you can now say why that can't work.

Vocabulary to own

- **t-test**: compares two means. **Chi-square / z-test for proportions**: compares two rates (most A/B tests). You don't need to compute them — know which is which.
- **Multiple comparisons**: test 20 metrics at $\alpha = 5 \%$ and one will be "significant" by luck. Pre-register the primary metric.
- **Novelty effect**: a new feature spikes then fades. Run long enough.
- **Shadow mode** (what you already use): the system decides, humans still decide, you compare — the safest first test for automations.

06 One-page cheat sheet

Base rate	How common the thing is before any evidence. Fraud ≈ 0.1 –0.5 %, default ≈ 2 –10 %. Everything downstream depends on it.
Bayes	posterior $\propto$ likelihood $\times$ prior. Rare events + imperfect detectors = most alerts are false. Hence precision.
$E[X]$, σ	Average and spread. Quote σ , not variance. $z = (x - \text{mean})/\sigma$ = "how unusual".
Heavy tails	Money is not normal. Medians and percentiles; log-transform.

CLT	Averages are normal; error shrinks with $\sqrt{n}$. 4× data → 2× precision.
95 % CI	mean $\pm 1.96 \cdot \sigma/\sqrt{n}$. Say the interval, not the point.
p-value	P(data this extreme no effect). Not P(no effect). Effect size + CI > p.
Power	Chance to detect a real effect; target 80 %. $n \approx 16\sigma^2/\delta^2$ per group.
Bias vs. noise	More data fixes noise, never bias. Survivorship bias in credit: you only see approved loans.
A/B hygiene	One primary metric, pre-set sample size, randomise by customer, full weeks, no peeking.

07 Ten questions — answer out loud, in English, then check

1. A model has 95 % accuracy on fraud detection. Why might that be useless? What would you ask for instead?
2. Fraud base rate 0.5 %; detector catches 95 % of fraud and false-alarms on 1 % of legit. A transaction is flagged: probability it's fraud?
3. Average customer balance is €2,300, median €410. What does that tell you, and which would you put on a dashboard?
4. You have 100 data points and a standard error of 5. How many points do you need to get the standard error to 2.5?
5. An A/B test shows $p = 0.03$ and a lift of 0.1 pp in conversion. Ship it?
6. Your credit model is trained only on loans that were approved. What's the problem, and what's it called?
7. Explain Type I and Type II error using a fraud alert as the example.
8. Why shouldn't you stop an A/B test the first day it shows significance?
9. Disputes per merchant per month: mean 4, variance 40. What might that suggest?
10. What's the difference between $P(\text{flag} | \text{fraud})$ and $P(\text{fraud} | \text{flag})$, and which one does the business care about?

Answers (check after, not before)

1. If fraud is 0.5 %, a model that says "never fraud" is 99.5 % accurate. Accuracy is meaningless with imbalanced classes. Ask for precision and recall (or PR-AUC) at the operating threshold, and the base rate.
2. $P(\text{flag}) = 0.95 \cdot 0.005 + 0.01 \cdot 0.995 = 0.00475 + 0.00995 = 0.0147$. $P(\text{fraud} | \text{flag}) = 0.00475 / 0.0147 \approx 32 \%$. Two out of three alerts are false.
3. Heavy right tail: a few very large balances drag the mean up; the typical customer has ~€410. Dashboard: median plus percentiles (p25/p75/p95); mean only for totals.
4. Error shrinks with $\sqrt{n}$. Halving it needs 4× the data: **400 points**.
5. Probably not: significant but tiny. Ask: is 0.1 pp worth the cost and complexity? What's the CI — does it include zero? Were there multiple metrics tested? Effect size decides, not p.
6. **Survivorship / selection bias**: you never observe what rejected applicants would have done, so the model learns from a biased population and can't evaluate its own rejections. Mitigations: reject inference, small random approval samples, external data.
7. Type I = flagging a legitimate transaction as fraud (false positive; annoys a customer). Type II = missing a real fraud (false negative; loses money). The threshold trades one for the other.

8. With repeated peeking the chance of a false “significant” somewhere along the way climbs far above 5 %. Fix the sample size in advance (or use sequential-testing methods designed for peeking).
9. Variance $\gg$ mean means the counts are not Poisson — they're over-dispersed: disputes are clustered on some merchants or some periods. Look for a ring, a broken merchant, or a bug.
10. $P(\text{flag} | \text{fraud})$ is recall (sensitivity): of real frauds, how many we catch. $P(\text{fraud} | \text{flag})$ is precision: of alerts, how many are real. Business cares about *both*, but ops teams live on precision (how many alerts are wasted) and losses depend on recall.

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Next: Module 2 – What a model is: regression, overfitting, regularisation, and the metrics (precision, recall, ROC / PR-AUC, calibration).